The Dual Meixner–Pollaczek Matching Coefficients: the Calculation

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1 The Problem

Let P_n be the Meixner–Pollaczek polynomials ($\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$),

$$(n+1)P_{n+1} = 2xP_n - (n - \tfrac{1}{2})P_{n-1}, \qquad P_0 = 1, \quad P_1 = 2x,$$

orthogonal for the weight $W(x) = \Gamma(\frac{1}{4} + ix)\,\Gamma(\frac{1}{4} - ix)$ on $\mathbb{R}$, with $\int P_m P_k W = h_k \delta_{mk}$, $h_0 = \sqrt{2}\,\pi^{3/2}$, and $\widehat{P}_n = P_n/\sqrt{h_n}$.

The weight factors into an analytic function and its reflection, and distributing one factor to each side defines a *pair of dual Meixner–Pollaczek functions*

$$F_m(x) = \Gamma(\tfrac{1}{4} + ix)\,\widehat{P}_m(x), \qquad G_m(x) = \Gamma(\tfrac{1}{4} - ix)\,\widehat{P}_m(x),$$

biorthonormal under the bilinear pairing: $\int_{\mathbb{R}} F_m G_k\,dx = \delta_{mk}$.

Problem. Approximate each dual function by Meixner–Pollaczek polynomials: find the coefficients in the best $L^2(W)$-approximation

$$F_m \approx \sum_{k \le N} c_{mk}\,\widehat{P}_k, \qquad c_{mk} = \langle F_m, \widehat{P}_k \rangle_W = \frac{J_{mk}}{\sqrt{h_m h_k}},$$

where

$$J_{mk} = \int_{\mathbb{R}} P_m(x)\,P_k(x)\,\Gamma(\tfrac{1}{4} + ix)^2\,\Gamma(\tfrac{1}{4} - ix)\,dx. \tag{1}$$

The coefficients for G_m are the complex conjugates, so it suffices to compute J_{mk}. We do so in closed form.

2 The Three-Gamma Integral

Proposition 2.1. *For* $\operatorname{Re} a, b, c > 0$,

$$\int_{\mathbb{R}} \Gamma(a+ix)\,\Gamma(b+ix)\,\Gamma(c-ix)\,dx = 2\pi\,\Gamma(a+c)\,\Gamma(b+c)\,U(a+c,\,a-b+1,\,1),$$

where U is Tricomi's confluent hypergeometric function.

Proof. Write each factor as $\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt$ with variables t, u (for $+ix$) and v (for $-ix$). The x-integral gives $\int_{\mathbb{R}} (tu/v)^{ix} dx = 2\pi\,\delta(\log(tu/v))$, which collapses the v-integral at $v = tu$:

$$\int_{\mathbb{R}} = 2\pi \int_0^\infty \int_0^\infty t^{a+c-1}\,u^{b+c-1}\,e^{-t-u-tu}\,dt\,du.$$

The u-integral is $\Gamma(b+c)(1+t)^{-(b+c)}$, and the remaining t-integral is Tricomi's representation $\Gamma(\alpha)\,U(\alpha,\beta,1) = \int_0^\infty t^{\alpha-1}(1+t)^{\beta-\alpha-1}e^{-t}\,dt$ with $\alpha = a+c$, $\beta = a-b+1$. □

The same proof with an oscillatory factor e^{isx} inserts e^{-s} into the delta constraint ($v = tue^{-s}$) and gives the shifted form

$$\int_{\mathbb{R}} \Gamma(a+ix)\,\Gamma(b+ix)\,\Gamma(c-ix)\,e^{isx}\,dx = 2\pi\,e^{-as}\,\Gamma(a+c)\,\Gamma(b+c)\,U(a+c,\,a-b+1,\,e^{-s}). \quad (2)$$

At $a = b = c = \frac{1}{4}$, using $U(\frac{1}{2}, 1, 2z) = e^z K_0(z)/\sqrt{\pi}$,

$$J_{00} = 2\pi\,\Gamma(\tfrac{1}{2})^2\,U(\tfrac{1}{2}, 1, 1) = 2\pi^{3/2}\sqrt{e}\;K_0(\tfrac{1}{2}) = 16.97348192607767\ldots \quad (3)$$

(verified to 20 digits by quadrature).

3 Reduction of the Polynomial Moments

The recurrence $(a+ix)\,\Gamma(a+ix) = \Gamma(a+1+ix)$ converts multiplication by x into integer shifts of the parameters:

$$ix\,\Gamma(a+ix) = \Gamma(a+1+ix) - a\,\Gamma(a+ix), \qquad -ix\,\Gamma(c-ix) = \Gamma(c+1-ix) - c\,\Gamma(c-ix).$$

Applying these repeatedly, every moment $\int x^j\,\Gamma(\frac{1}{4}+ix)^2\Gamma(\frac{1}{4}-ix)\,dx$ is a $\mathbb{Q}$-combination of integrals of Proposition 2.1 with parameters in $\frac{1}{4}+\mathbb{N}$, hence of values $U(\alpha, \beta, 1)$ with $\alpha \in \frac{1}{2}+\mathbb{N}$, $\beta \in \mathbb{Z}$. The contiguous relations for U reduce all of these to the $\mathbb{Q}$-span of two generators, which we take (via $K_\nu(z) = \sqrt{\pi}\,(2z)^\nu e^{-z}\,U(\nu+\frac{1}{2}, 2\nu+1, 2z)$) to be

$$A = \sqrt{2e}\;K_0(\tfrac{1}{2}) = 2.155416163\ldots, \qquad B = \sqrt{2e}\;K_1(\tfrac{1}{2}) = 3.862230968\ldots$$

Since the P_m have rational coefficients, every J_{mk}/h_0 lies in $\mathbb{Q}A \oplus \mathbb{Q}B$. The denominators are exactly $2^m m!\,2^k k!$.

Theorem 3.1.

$$T_{mk} = \frac{2^m m!\,2^k k!}{h_0}\,J_{mk} \;\in\; \mathbb{Z}A \oplus \mathbb{Z}B \qquad (T_{mk} \textit{ real for } m+k \textit{ even, } i\cdot\textit{real for } m+k \textit{ odd}). \quad (4)$$

The proof is a closed combinatorial formula for the two integer matrices, extracted from the generating function of §4; see Theorem 4.3.

4 The Generating Function

The whole table has a closed generating function. Substituting the Meixner–Pollaczek generating function

$$\sum_{n\ge 0} P_n(x)\,w^n = (1-iw)^{-\frac{1}{4}+ix}(1+iw)^{-\frac{1}{4}-ix}$$

once in each index of (1) turns all x-dependence into a single oscillation e^{isx} with

$$e^{-s} = \zeta(w,u) = \varphi(w)\,\varphi(u), \qquad \varphi(t) = \frac{1+it}{1-it} \quad \text{(the Cayley transform)},$$

and (2) evaluates the integral. The algebraic prefactors collapse, and using $U(\frac{1}{2}, 1, 2z) = e^z K_0(z)/\sqrt{\pi}$ once more,

$$\sum_{m,k\ge 0} J_{mk}\,w^m u^k = \frac{2\pi^{3/2}\,e^{\zeta/2}\,K_0(\zeta/2)}{\sqrt{(1-iw)(1-iu)}} \quad (5)$$

(verified numerically to 15 digits against every entry of the table). At $w = u = 0$, $\zeta = 1$, recovering (3).

Formula (5) explains the two-generator structure: every Taylor coefficient acts on K_0 through the closed derivative algebra $K_0' = -K_1$, $K_1' = -K_0 - K_1/z$, evaluated at $z = \frac{1}{2}$, so each J_{mk}/h_0 lies in $\mathbb{Q}A \oplus \mathbb{Q}B$ automatically (proving what §3 stated empirically, up to the denominators). It also makes the real/imaginary checkerboard of (4) visible by inspection: $(w, u) \mapsto (-w, -u)$ conjugates ζ.

The derivative algebra is a weighted *two-state automaton* on states $\{K_0, K_1\}$: each differentiation either switches state (weight -1) or loops at K_1 (weight $-1/z$, i.e. -2 at $z = \frac{1}{2}$). Expanding (5) by Faà di Bruno, the integer coefficients of A and B in T_{mk} are thus signed sums over pairs (set partition of the $m+k$ derivatives, path in the automaton with one step per block), each derivative dressed by a Taylor coefficient of $\zeta = \varphi(w)\varphi(u)$ at the Cayley transform — the source of the powers of 2 absorbed by the normalisation $2^m m!\, 2^k k!$ in (4). We now extract the self-contained formula, and with it the proof of Theorem 3.1.

The combinatorial formula

The substitution that trivialised the reverse direction (§6) also un-nests the forward generating function. In the rescaled variables $v = 2iw$, $\mu = 2iu$ set

$$\sigma = \frac{v}{1 - v/2}, \qquad \tau = \frac{\mu}{1 - \mu/2}.$$

Proposition 4.1. *$\zeta = \varphi(w)\,\varphi(u) = (1+\sigma)(1+\tau)$, that is, $\zeta - 1 = \sigma + \tau + \sigma\tau$, and the algebraic prefactor of (5) is $\big(\sigma'(v)\,\tau'(\mu)\big)^{1/4}$. Hence*

$$\sum_{m,k \ge 0} J_{mk}\, w^m u^k = 2\pi^{3/2} \big(\sigma'\tau'\big)^{1/4}\, g\big((1+\sigma)(1+\tau)\big), \qquad g(\zeta) = e^{\zeta/2} K_0(\zeta/2).$$

Proof. $1 + \sigma = (1 + v/2)/(1 - v/2) = (1 + iw)/(1 - iw) = \varphi(w)$, and $\sigma'(v) = (1 - v/2)^{-2} = (1 - iw)^{-2}$. □

This is the exact mirror of the reverse kernel of §6, which factors through $\sigma = v/(1 + v/2)$ as $e^{\sigma+\tau-\sigma\tau}$; the forward direction replaces the exponential by the Bessel kernel g and flips the sign of the interaction term. The Taylor coefficients of g at the base point $\zeta = 1$ are controlled by Kummer's equation.

Lemma 4.2. *$g(\zeta) = \sqrt{\pi}\, U(\frac{1}{2}, 1, \zeta)$ satisfies $\zeta g'' + (1 - \zeta)g' - \frac{1}{2}g = 0$. Consequently the rescaled Taylor coefficients $u_n = 2^n\sqrt{2}\; g^{(n)}(1)$ obey the integer recurrence*

$$u_{n+2} = -2n\, u_{n+1} + (4n + 2)\, u_n, \qquad u_0 = A, \quad u_1 = A - B,$$

so that every u_n lies in $\mathbb{Z}A \oplus \mathbb{Z}B$:

$$u_n = A,\ A - B,\ 2A,\ 2A - 6B,\ 12A + 24B,\ -44A - 228B,\ 568A + 2256B,\ \ldots$$

Proof. The first claim is Kummer's equation for $U(a, b, \zeta)$ at $a = \frac{1}{2}$, $b = 1$. Differentiating it n times and setting $\zeta = 1$ gives $g^{(n+2)} + n\, g^{(n+1)} - (n + \frac{1}{2})\, g^{(n)} = 0$; the rescaling by 2^n clears the half-integer. The initial values are $\sqrt{2}\, g(1) = \sqrt{2e}\, K_0(\frac{1}{2}) = A$ and $2\sqrt{2}\, g'(1) = \sqrt{2e}\,(K_0 - K_1)(\frac{1}{2}) = A - B$, using $g' = \frac{1}{2}e^{\zeta/2}(K_0 - K_1)(\zeta/2)$. □

Theorem 4.3. *With $L(n, c) = \binom{n-1}{c-1}\, n!/c!$ the Lah numbers and $(2d-1)!! = 1 \cdot 3 \cdots (2d-1)$,*

$$T_{mk} = i^{\,m+k} \sum_{d_1, c_1, d_2, c_2} V_m(d_1, c_1)\; V_k(d_2, c_2) \sum_{j \ge 0} \binom{c_1}{j} \binom{c_2}{j}\, j!\; 2^j\; u_{c_1 + c_2 - j},$$

$$V_n(d, c) = \binom{n}{d}\, (2d-1)!!\; L(n - d,\, c)\; 2^{\,n-d}.$$

Every weight is a positive integer, so $T_{mk} \in \mathbb{Z}[i]\, A \oplus \mathbb{Z}[i]\, B$, and the phase i^{m+k} produces the real/imaginary checkerboard of (4). This proves Theorem 3.1.

Proof. Expand g around $\zeta = 1$ in Proposition 4.1 and apply the multinomial theorem to $(\sigma+\tau+\sigma\tau)^n$ with a factors σ, b factors τ and j factors $\sigma\tau$; write $c_1 = a+j$, $c_2 = b+j$, so $n = c_1+c_2-j$ and $1/(a!\,b!\,j!) = \binom{c_1}{j}\binom{c_2}{j} j!/(c_1!\,c_2!)$. From $\sigma^{c_1} = (2iw)^{c_1}(1-iw)^{-c_1}$,

$$[w^{m'}]\,\sigma^{c_1} = 2^{c_1}\, i^{\,m'} \binom{m'-1}{c_1-1},$$

and for the prefactor

$$2^d\, d!\; [w^d]\,(1-iw)^{-1/2} = i^{\,d}\,\frac{(2d)!}{2^d\, d!} = i^{\,d}\,(2d-1)!!,$$

where the power of 2 balances exactly by Legendre's formula. All phases combine into the single factor $i^{\,d_1+m'+d_2+k'} = i^{\,m+k}$. Splitting $m = d_1 + m'$ against the prefactor, writing $m! = \binom{m}{d_1}\, d_1!\, m'!$ and $m'!\,\binom{m'-1}{c_1-1}/c_1! = L(m', c_1)$ yields $V_m(d_1, c_1)$, with 2^m distributed as $2^{d_1}\cdot 2^{m'}$; the surplus $2^{c_1+c_2}$ from the chains against 2^{-n} in $u_n = 2^n\sqrt{2}\, g^{(n)}(1)$ leaves the matching weight 2^j. The formula was verified against the exact expansion of the generating function for all $m, k \le 6$, and the expansion itself against 40-digit quadrature of (1) (`mp_symbolic_lattice.py`). □

The comparison with §6 is now sharp. The reverse kernel $e^{\sigma+\tau-\sigma\tau}$ carries all its signs in the combinatorial weights (defects $(-1)^d$, chains $(-1)^{\ell-1}$, matchings -1) and none in the analytic factor, which is constant; the forward kernel has all weights positive — defects $(2d-1)!!$, chains through the Lah numbers, matchings $2^j\, j!$ — and every sign of the table is carried by the Kummer sequence u_n alone. One Gamma factor gives a one-state (exponential) kernel and the single generator $2\pi/e$; three Gamma factors give the two-state Bessel kernel and the rank-two lattice $\mathbb{Z}A \oplus \mathbb{Z}B$.

5 The Coefficients

The table lists the rescaled integrals T_{mk} of (4), with integer coefficients computed exactly by symbolic expansion of the generating function (5) (the factor i for $m+k$ odd is dropped). The entries were originally recovered by PSLQ at 40 digits, which the exact expansion confirms; $(3,5)$, beyond the PSLQ coefficient search bound, is supplied by the symbolic computation. To recover the approximation coefficients of F_m: since $h_n = h_0\binom{2n}{n}/4^n$, the normalisations collapse to

$$c_{mk} = \frac{T_{mk}}{m!\, k!\,\sqrt{\binom{2m}{m}\binom{2k}{k}}},$$

and the coefficients of G_m are the complex conjugates.

(m,k)	T_{mk}	(m,k)	T_{mk}
$(0,0)$	A	$(1,3)$	$2575A - 1440B$
$(0,1)$	$3A - 2B$	$(1,4)$	$39411A - 21978B$
$(0,2)$	$12B - 23A$	$(1,5)$	$394932B - 707389A$
$(0,3)$	$138B - 241A$	$(2,2)$	$2369A - 1320B$
$(0,4)$	$3265A - 1800B$	$(2,3)$	$33743A - 18834B$
$(0,5)$	$52907A - 29706B$	$(2,4)$	$317892B - 569719A$
$(1,1)$	$12B - 21A$	$(2,5)$	$6190602B - 11092229A$
$(1,2)$	$114B - 205A$	$(3,3)$	$295572B - 529549A$
		$(3,4)$	$5399634B - 9675553A$
		$(3,5)$	$201409319A - 112403832B$

6 The Reverse Expansion

Expanding in the other direction — the polynomials in terms of the dual functions — is elementary by contrast. Biorthogonality gives

$$\widehat{P}_m = \sum_k D_{mk} F_k, \qquad D_{mk} = \int_{\mathbb{R}} \widehat{P}_m \, G_k \, dx = \frac{1}{\sqrt{h_m h_k}} \int_{\mathbb{R}} P_m(x)\, P_k(x)\, \Gamma(\tfrac{1}{4} - ix)\, dx,$$

a *one*-Gamma integral. Mellin inversion of $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ gives $\int_{\mathbb{R}} \Gamma(a+ix)\, t^{-ix} dx = 2\pi\, t^a e^{-t}$, and the same substitution of the generating function as in §4 yields

$$\sum_{m,k\geq 0} \Big(\int_{\mathbb{R}} P_m P_k \, \Gamma(\tfrac{1}{4} - ix)\, dx \Big) w^m u^k = \frac{2\pi\, e^{-1/\zeta}}{\sqrt{(1+iw)(1+iu)}} \tag{6}$$

(verified to 20 digits; at $w = u = 0$ this is $2\pi/e$). Where the forward direction needed the Bessel pair (A, B), the reverse needs the single generator e^{-1}: one Gamma factor inverts to a pure exponential, whose derivative algebra has one state, while three Gamma factors produce the Tricomi U, whose Bessel algebra has two. The two generating functions are exchanged by $\zeta \mapsto 1/\zeta$, $(w, u) \mapsto (-w, -u)$ — the Cayley inversion swapping the half-planes.

The same normalisation as in (4) again produces integers — now over the *single* generator $2\pi/e$, and provably so, since the Taylor coefficients of (6) are Gaussian rationals times e^{-1} and can be expanded symbolically (no PSLQ needed):

$$S_{mk} = 2^m m!\, 2^k k!\, \frac{e}{2\pi} \int_{\mathbb{R}} P_m P_k \, \Gamma(\tfrac{1}{4} - ix)\, dx \;\in\; \mathbb{Z}, \qquad D_{mk} = \frac{1}{e}\sqrt{\frac{2}{\pi}}\, \frac{S_{mk}}{m!\, k! \, \sqrt{\binom{2m}{m}\binom{2k}{k}}}$$

(again dropping a factor i when $m + k$ is odd).

S_{mk}	$k=0$	1	2	3	4	5
$m=0$	1	3	5	11	249	−2621
1		7	47	399	−2005	263
2			473	−617	22109	−402065
3				31511	−412557	−3017969
4					−1916623	−29235989
5						−1012892921

These integers have a combinatorial interpretation. In the rescaled variables $v = 2iw$, $\mu = 2iu$ the generating function (6) factors as

$$\big(\sigma'(v)\, \tau'(\mu)\big)^{1/4}\, e^{\sigma+\tau-\sigma\tau}, \qquad \sigma = \frac{v}{1+v/2}, \quad \tau = \frac{\mu}{1+\mu/2}$$

— a Charlier kernel in Laguerre variables, with the Meixner–Pollaczek parameter $\lambda = \frac{1}{4}$ appearing as the exponent of the Jacobian prefactor. Reading this as a species:

Proposition 6.1. *$S_{mk} = (-1)^{\lfloor (m+k)/2 \rfloor} \sum_{\text{structures}} \mathrm{wt}$, summed over structures on $([m], [k])$ in which each of $[m]$, $[k]$ is split into a defect set and a forest of nonempty chains (linear orders), together with a partial matching between the $[m]$-chains and the $[k]$-chains; the weights are $(-1)^d (2d-1)!!$ for a defect set of size d (the prefactor), $(-1)^{\ell-1}\, 2^{\ell+1}$ for a chain of length ℓ (the substitution), and -1 per matched pair of chains (the Charlier kernel $e^{\sigma+\tau-\sigma\tau}$).*

Equivalently, with Lah numbers $L(n, c) = \binom{n-1}{c-1} n!/c!$ counting the chain forests,

$$S_{mk} = (-1)^{\lfloor \frac{m+k}{2} \rfloor} \sum_{d_1, c_1, d_2, c_2} W_m(d_1, c_1)\, W_k(d_2, c_2)\, C_{c_1 c_2},$$

$$W_n(d,c) = \binom{n}{d}(-1)^d(2d-1)!!\; L(n{-}d,\, c)\,(-1)^{n-d-c}\,2^{n-d+c},$$

where $C_{c_1c_2} = \sum_j (-1)^j \binom{c_1}{j}\binom{c_2}{j} j!$ is the signed count of partial matchings (a Charlier polynomial value). This reproduces the entire table exactly.

One caution: the coefficients D_{mk} *grow* along the diagonal ($D_{00} = 0.2935\ldots$, $D_{11} = 1.027\ldots$, $D_{22} = 5.785\ldots$), so the reverse expansion is formal — $\widehat{P}_m$ does not lie in the closed span of the F_k. The analytic content lives in the forward direction; the reverse carries the simpler arithmetic.